SMART FABRICS: The Comfortable Way To Wear Your Tech

Janani Gopalakrishnan Vikram is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

Wearable devices are the rage today, and these are getting better day by day—lighter, cooler, trendier, smarter. Nevertheless, it can be a pain wearing a chunky watch on a hot summer day. The fewer devices you have to wear, the more comfortable you will feel. Not to forget the fact that you may forget to wear your smartwatch or fitness monitor before you step out of the house.

For those who are more focused on fashion than comfort, once again devices fail to please. You might all remember how a few years ago Apple tried to make its Apple Watch a style statement—people like Raf Simons of Dior and Anna Wintour of Vogue were seen wearing it. But the trend soon changed. Nobody expressed the reason as precisely as writer Vanessa Friedman, in *The New York Times*, "No matter how attractive Apple Watch is in the context of other smartwatches or smartbands, no matter how much of an aesthetic advance its rounded corners and rectangular display, it still looks like a gadget." Is there a way out, to wear your technology without it being so obvious?

Fig. 1: Ohio State University researchers have developed a way of embroidering electronic components into clothing (Image courtesy: Ohio State University)

Smart fabrics is what we are getting at. These might not be as well-known as smartwatches, bracelets and pendants, but these are real, and one day in the near future, you might find these in your neighbourhood apparel stores and sports shops. According to a report by ABI Research, the smart clothing market will top 18 million clothing articles annually by 2021, representing a 48 per cent compound annual growth rate (CAGR).

Smart fabrics, or clothing into which technology is embedded or woven, is proving to be a boon for athletes, fitness lovers, first responders like firemen and policemen, people in need of special health care and others. Such garments range from jackets to underwear, which can monitor health parameters, activity, location and more. Together with mobile apps, these can help athletes train and normal people get back into shape, keep kids safe and elders comfortable.

It is not all about functionality either. Some of the world's greatest fashion designers are now using novel materials embedded with technology, for purely aesthetic purposes—like clothes that change colour or brightness according to the environment. The realisation that electronic/smart textiles need not necessarily be wearable also helps widen our view of possibilities. These fabrics can, for example, be used to make smart curtains, car seat covers or bed spreads for children or patients.

It is now becoming increasingly practicable to develop and manufacture such smart fabrics, thanks to developments in materials and nano technologies. We have super-strong, technology-conducive materials like spider silk, flexible batteries and electronics that can power themselves from the body's heat or vibrations. These are all contributing directly and indirectly to the field of smart fabrics. Let us take a look at how smart garments are today, and what makes these so.